MORE ABOUT ME

Personal + Interview Preparation Profile

1. Basic Personal Information

Full Name: Siddharth Yadav

Date of Birth: 25 December 2005

Current Course: B.Tech – Health Informatics (currently in 7th semester)

College: VIT Bhopal University

Based in: Kanpur, Uttar Pradesh

Interests: Data Science, Artificial Intelligence, Healthcare Technology

2. Hobbies & Interests

- **Football:** Playing/watching football; interested in teamwork, strategy and competition.
- **Gym & Fitness:** Regular strength training and maintaining a disciplined routine.
- **Technology:** Exploring AI, machine learning, data science and practical technology projects.
- **Learning:** Building programming and problem-solving skills through hands-on practice.

3. Goals

Short-term goal: Start my career in a fresher role in data analytics, data science, AI/ML or healthcare analytics, and keep building practical projects.

Long-term goal: To become a strong technology professional with practical experience in data, AI and problem-solving, while continuing to develop both technical and communication skills.

4. Internship at HAL: Deep Learning and OpenCV-Based Aircraft Component Identification

- **Organisation:** Hindustan Aeronautics Limited (HAL), Transport Aircraft Division (TAD), Kanpur, IT Department. HAL is India's aerospace and defence company, under the Ministry of Defence. TAD Kanpur was set up in 1960 and makes aircraft and aerospace structural assemblies, for example the HS-748, Dornier-228 and HPT-32. The division's IT Department supports its operations, including maintenance (MRO) workflow systems.
- **Period:** 9 May 2026 to 4 July 2026 (8 weeks), project training done as part of the B.Tech degree, under the guidance of the Chief Manager (IT) at HAL TAD Kanpur.

- **Team:** This was a team project. Siddharth worked with four other B.Tech students from different universities. The report describes what the team built together and does not list who did which part, so describe the work as the team's work unless Siddharth says otherwise.
- **Problem:** During Maintenance, Repair and Overhaul (MRO), aircraft are taken apart and many parts are stored together. Many structural parts (brackets, frames, ribs, levers, fairings) exist as mirror-image pairs: one for the left-hand side (LHS) of the aircraft and one for the right-hand side (RHS). They are geometrically identical except for orientation, so it is hard to tell them apart by eye without comparing them side by side or checking manuals. This risks wrong re-assembly (a critical safety issue in defence aviation), wastes time, allows human error and delays aircraft turnaround.
- **Objective:** Build an offline computer vision system that takes a photograph of a disassembled component and classifies it as LHS or RHS. It had to run completely offline on the HAL IT Department system with no internet dependency, and be accurate and robust enough for a defence MRO environment.
- **Approach:** A supervised deep learning pipeline for binary image classification (LHS vs RHS) using transfer learning with a pretrained ResNet-50, with a real-time webcam demo built with OpenCV.
- **Dataset:** Images of aircraft components collected and annotated at HAL Kanpur (the component images were provided by the HAL TAD IT Department), organised into LHS and RHS folders. Each image has a Pascal VOC XML annotation (class label plus bounding box), made with the LabelImg tool. The report describes the dataset as relatively limited and does not state how many images it contained.
- **Preprocessing:** The XML annotations were parsed with Python's ElementTree. Each component was cropped from the photo using its bounding box, so the model focuses on the part and not the background. Images were converted to RGB, resized to 224 x 224 pixels, turned into tensors and normalised with ImageNet mean and standard deviation (Pillow was used for image handling).
- **Data augmentation:** Random rotation (10 degrees), random resized crop and colour jitter (brightness and contrast), to make the model robust to orientation, scale and lighting changes.
- **Model:** Pretrained ResNet-50 (ImageNet weights) used as a feature extractor with its pretrained layers frozen. The final fully connected layer was replaced by a new 2048-to-2 layer for the two classes (LHS, RHS), and only that layer was trained. Transfer learning was chosen because it trains faster and works better than training from scratch on a limited dataset.
- **Training:** Custom PyTorch Dataset and DataLoader; 80/20 train/validation split; Adam optimiser with learning rate 0.001; CrossEntropy loss; 25 epochs, with loss monitored each epoch; trained weights saved as lhs_rhs_model.pth.
- **Real-time inference:** An OpenCV application captures webcam frames, converts them to RGB and PIL images, applies the same preprocessing as in training, runs the trained ResNet-

50, picks the class with the highest score and draws the label (LHS or RHS) on the live video. It runs fully offline.

- **Result:** A working proof-of-concept prototype that classifies components as LHS or RHS in real time from a webcam. It was demonstrated to the Head of Department at HAL Kanpur. The report includes sample live-video frames showing LHS and RHS predictions.
- **Accuracy figures:** The report does not record numeric accuracy, precision, recall, F1 or confusion-matrix results. The prototype was evaluated through the live demonstration. Adding quantitative evaluation is listed as future work.
- **Tools and technologies:** Python, PyTorch, torchvision (ResNet-50), OpenCV, Pillow, XML parsing with ElementTree, LabelImg for annotation.
- **Future scope (from the report):** Add an object detector (such as YOLO or Faster R-CNN) to locate and crop the component before classifying it, because training used cropped components while the live demo classifies the whole webcam frame; fine-tune more of ResNet-50 using a larger dataset; collect more images under different orientations, lighting and backgrounds; extend to multi-class component classification; add quantitative evaluation metrics; deploy on edge devices or industrial workstations; convert the model to TorchScript or ONNX for mobile use.
- **Why there is no code or GitHub link:** HAL is a defence organisation and did not allow the students to take or share the code, so this project has no public repository. The work is documented in the project report submitted to HAL.

5. Interview Questions and My Answers

Tell me about yourself.

I am Siddharth Yadav, a B.Tech Health Informatics student at VIT Bhopal University, currently in my 7th semester. I am interested in using data and AI to solve healthcare problems. I have built three end-to-end projects: a machine learning pipeline that predicts hospital readmission risk for diabetic patients, an AI tool that compares a resume against a job description, and an AI agent that answers questions about a real estate property. I also completed an internship at Hindustan Aeronautics Limited (HAL) in summer 2026.

Walk me through your background.

I am studying B.Tech in Health Informatics at VIT Bhopal, with coursework in data structures and algorithms, databases, AI, machine learning, data analytics, statistics and electronic health records. Alongside my degree I did an internship and summer training in Deep Learning and OpenCV at HAL, and completed certifications including Applied Machine Learning from the University of Michigan and Generative AI using IBM watsonx. I then applied this through three projects, which are all on my GitHub.

Why did you choose your degree/branch?

I was interested in the intersection of healthcare and technology. Health Informatics lets me work with data, AI and technology while solving problems connected to healthcare.

Why do you want this role?

I want a role where I can work with data and AI on practical problems, ideally connected to healthcare, and keep learning from experienced people while contributing from the start.

Why should we hire you?

I have built three end-to-end projects on my own, from data cleaning and modelling to a working interface, so I already know how to take a problem from start to finish. I am consistent, I keep working on a difficult topic until I understand it, and I am willing to learn whatever the role needs.

What are your strengths?

Consistency, willingness to learn, problem-solving, and the ability to keep working on a difficult topic until I understand it.

What is one area you are improving?

I am continuously improving my DSA and programming fundamentals. I focus on understanding the logic rather than only memorizing solutions.

Tell me about a project you are proud of.

My Healthcare Patient Readmission Prediction project. I built an end-to-end pipeline on the Diabetes 130-US Hospitals dataset to predict 30-day readmission risk. It covered data cleaning, exploratory analysis, feature engineering, Random Forest and XGBoost models, and SHAP to explain the predictions, plus a Power BI dashboard concept and a Streamlit interface. It combines healthcare, data and machine learning, which is what I want to work on.

What motivates you?

Turning data into useful insights and practical solutions, especially in healthcare.

What did you do at HAL?

At the Transport Aircraft Division of HAL in Kanpur, I was part of a team of five students who built a deep learning system to tell left-hand and right-hand mirror-image aircraft parts apart from a photo. During maintenance and overhaul, many such parts are stored together and are hard to tell apart by eye, which can lead to wrong re-assembly. We trained a ResNet-50 model with transfer learning on annotated component images and built a real-time OpenCV webcam application that shows LHS or RHS on the live video. It ran fully offline, and we demonstrated it to the Head of Department at HAL Kanpur.

Why is there no code or GitHub link for the HAL project?

HAL is a defence organisation and did not allow us to take or share the code, so I can't publish it. The work is documented in the project report we submitted to HAL, and I can explain the approach, the model and the pipeline in detail.

Why are you interested in Data Science / AI?

I enjoy working with data and solving problems logically. I am particularly interested in using machine learning and AI to turn data into useful insights and practical solutions.

What do you do when you get stuck on a problem?

I first break the problem into smaller parts, learn the missing concept, practice it, and then apply it to the original problem.

How do you handle something you don't know?

I first break the problem into smaller parts, learn the missing concept, practice it, and then apply it to the original problem.

What do you do outside academics?

I spend time playing or following football, going to the gym, and learning technical skills through coding and projects.

Where do you see yourself in 3–5 years?

I want to become a strong technology professional with practical experience in data, AI and problem-solving, while continuing to develop both technical and communication skills.